Computer Architecture

Lecture 2: Fundamental Concepts and ISA

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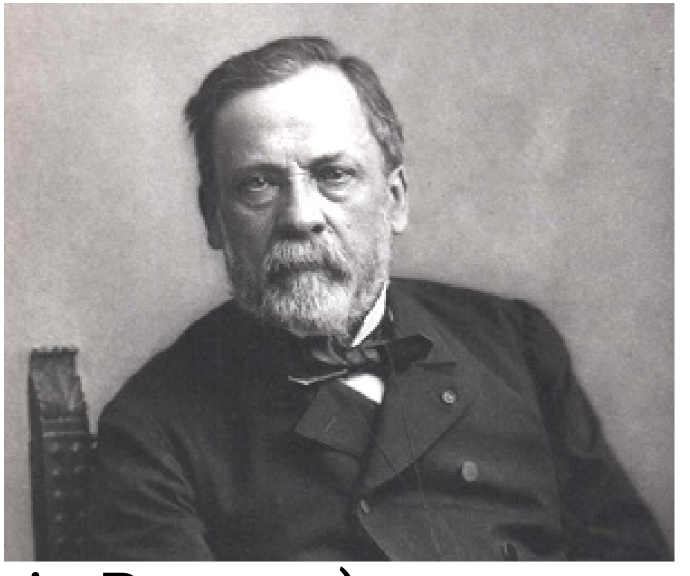
Based on original slides by Prof. Onur Mutlu

What Do I Expect From You?

“Chance favors the prepared mind.”



(Louis Pasteur)

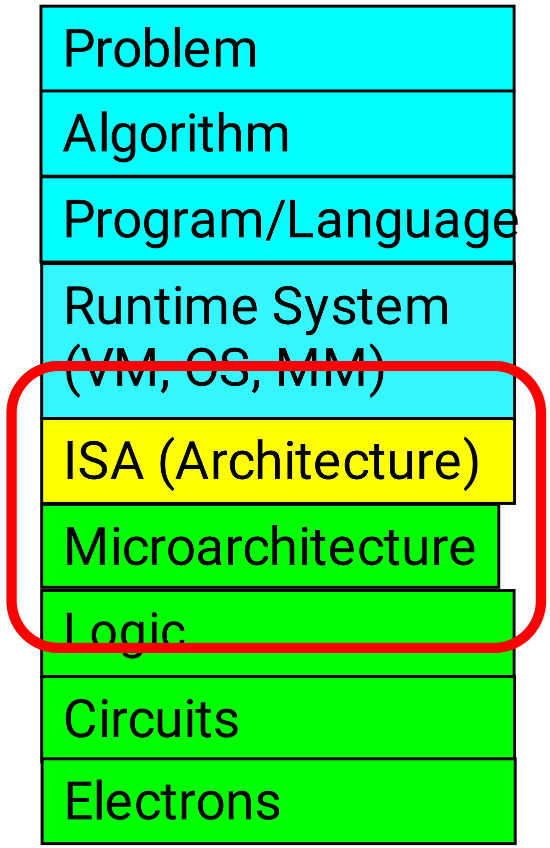
“كل عامل ميسر لعمله”(محمد، صلي اللّه عليه وسلم)

Agenda for Today

- Review
- Why study computer architecture?
- ISA
- Microarchitecture
- Computer Architecture
- Next Lecture

REVIEW

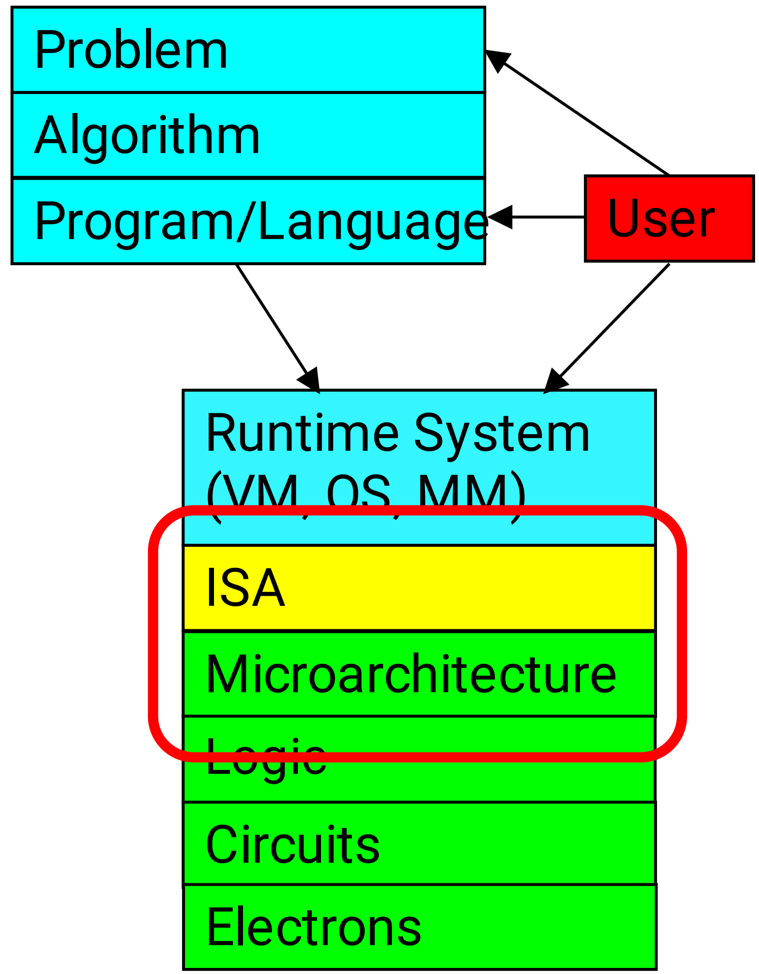
Review: Comp. Arch. in Levels of Transformation



- Read: Patt, “Requirements, Bottlenecks, and Good Fortune: Agents for Microprocessor Evolution,” Proceedings of the IEEE 2001.

Review: Levels of Transformation, Revisited

- A user-centric view: computer designed for users



- The entire stack should be optimized for user

Course Goals

- Goal 1: To familiarize those interested in computer system design with both fundamental operation principles and design tradeoffs of processor, memory, and platform architectures in today's systems.
  - Strong emphasis on fundamentals and design tradeoffs.

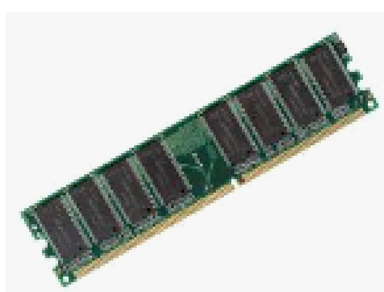
- Goal 2: To provide the necessary background and experience to design, implement, and evaluate a modern processor by performing hands-on RTL and C-level implementation.
  - Strong emphasis on functionality and hands-on design.

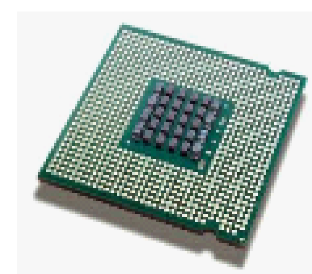
- Goal 3: Think critically (in solving problems), and broadly across the levels of transformation

WHY STUDY COMPUTER ARCHITECTURE?

Why study Computer Architecture?

- The science and art of designing, selecting, and interconnecting hardware components and designing the hardware/software interface to create a computing system that meets functional, performance, energy consumption, cost, and other specific go

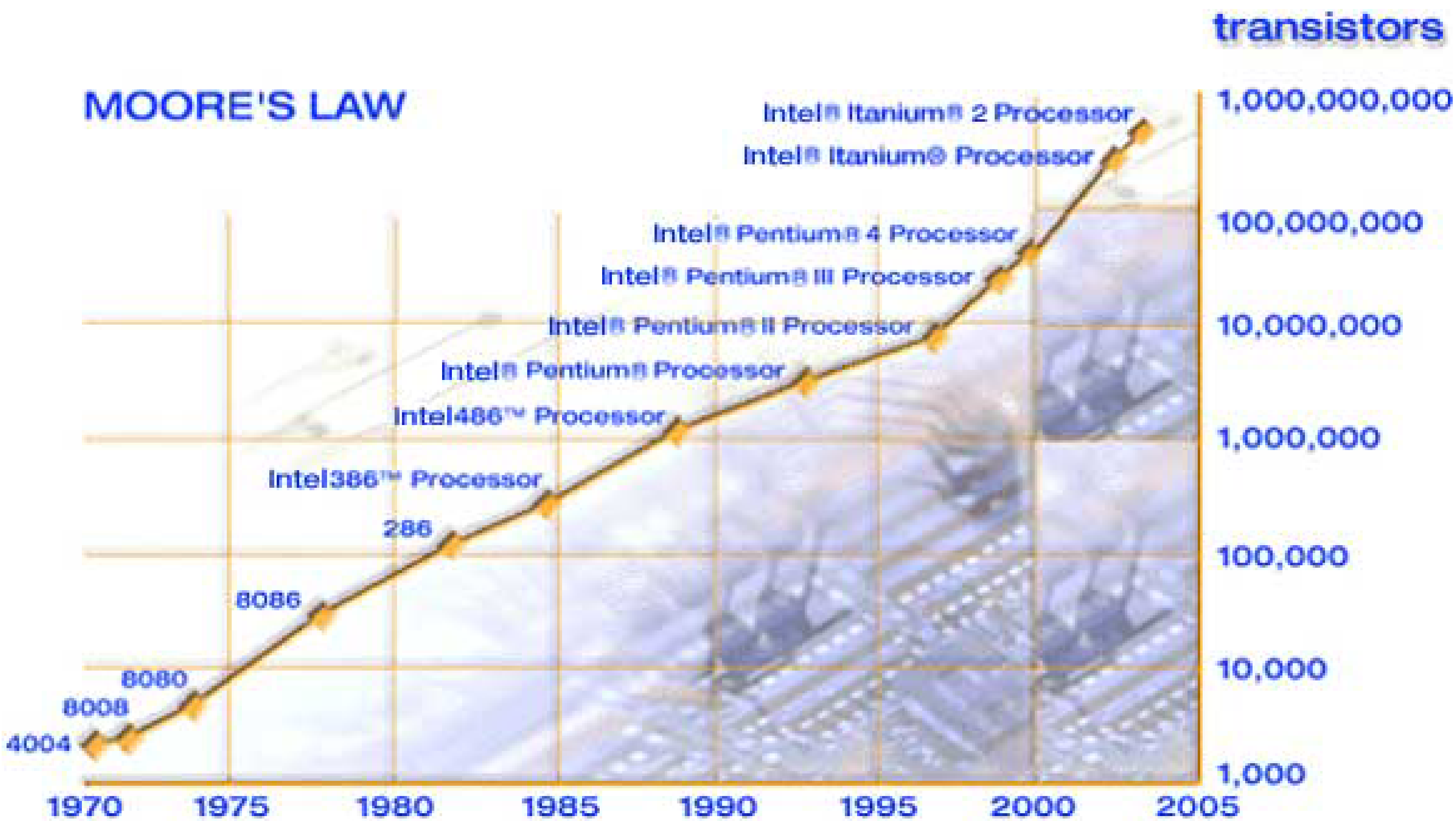




- We will soon distinguish between the terms *architecture*, and *microarchitecture*.

إن شاء الله

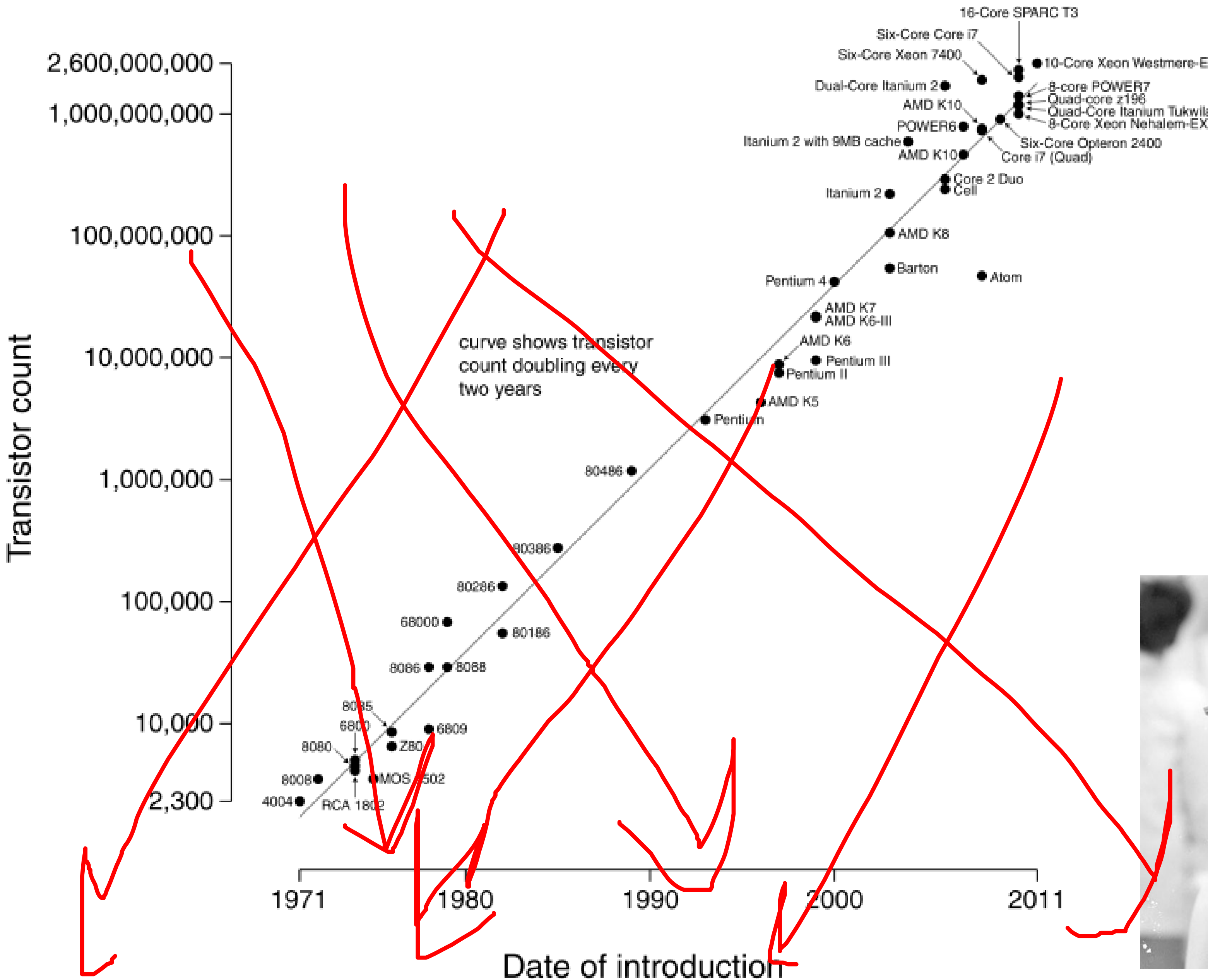
An Enabler: Moore's Law

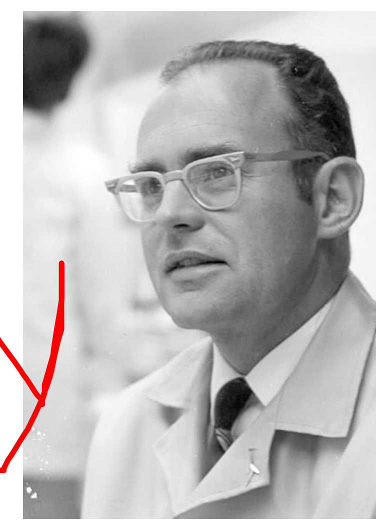


Moore, "Cramming more components onto integrated circuits," Electronics Magazine, 1965. Component counts double every other year

Image source: Intel

Microprocessor Transistor Counts 1971-2011 & Moore's Law





Number of transistors on an integrated circuit doubles ~ every two years

Image source: Wikipedia

What Do We Use These Transistors for?

- Your readings for this week should give you an idea…

- Patt, “Requirements, Bottlenecks, and Good Fortune: Agents for Microprocessor Evolution,” Proceedings of the IEEE 2001.

Computer Architecture Today

- Today is a very exciting time to study computer architecture

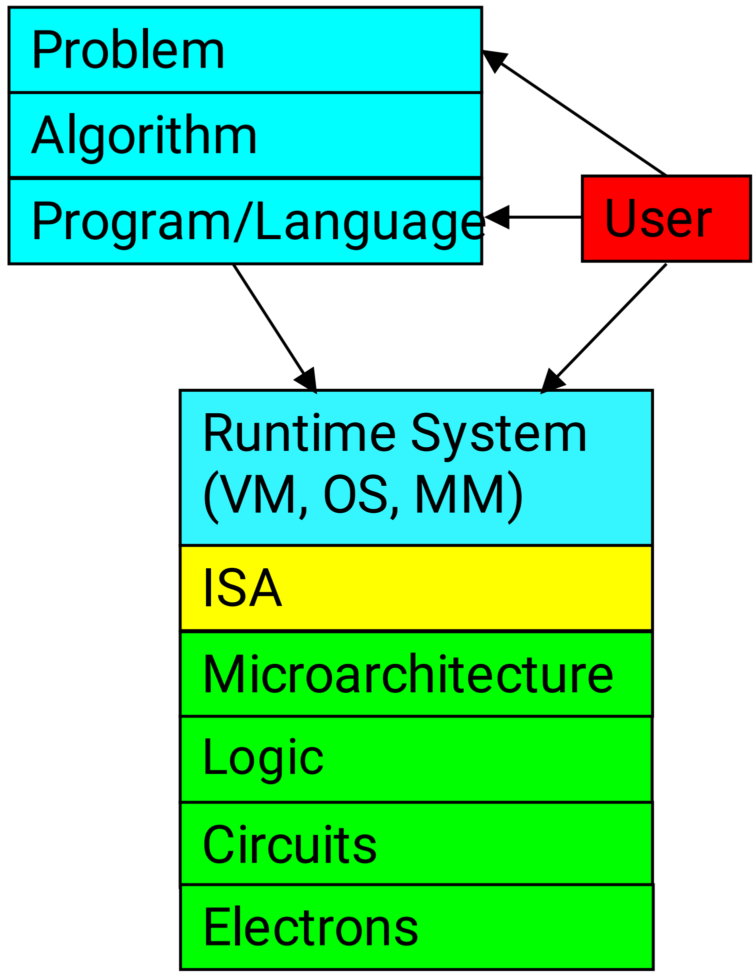
- Industry is in a large paradigm shift (to multi-core and beyond) – many different potential system designs possible

- Many difficult problems *motivating* and *caused by* the shift
  - Power/energy constraints
  - Complexity of design → multi-core?
  - Difficulties in technology scaling → new technologies?
    - Memory wall/gap
    - Reliability wall/issues
    - Programmability wall/problem

- No clear, definitive answers to these problems

Computer Architecture Today (II)

- These problems affect all parts of the computing stack – if we do not change the way we design systems



- You can invent new paradigms for computation, communication, and storage

… but, first …

- Let's understand the fundamentals…

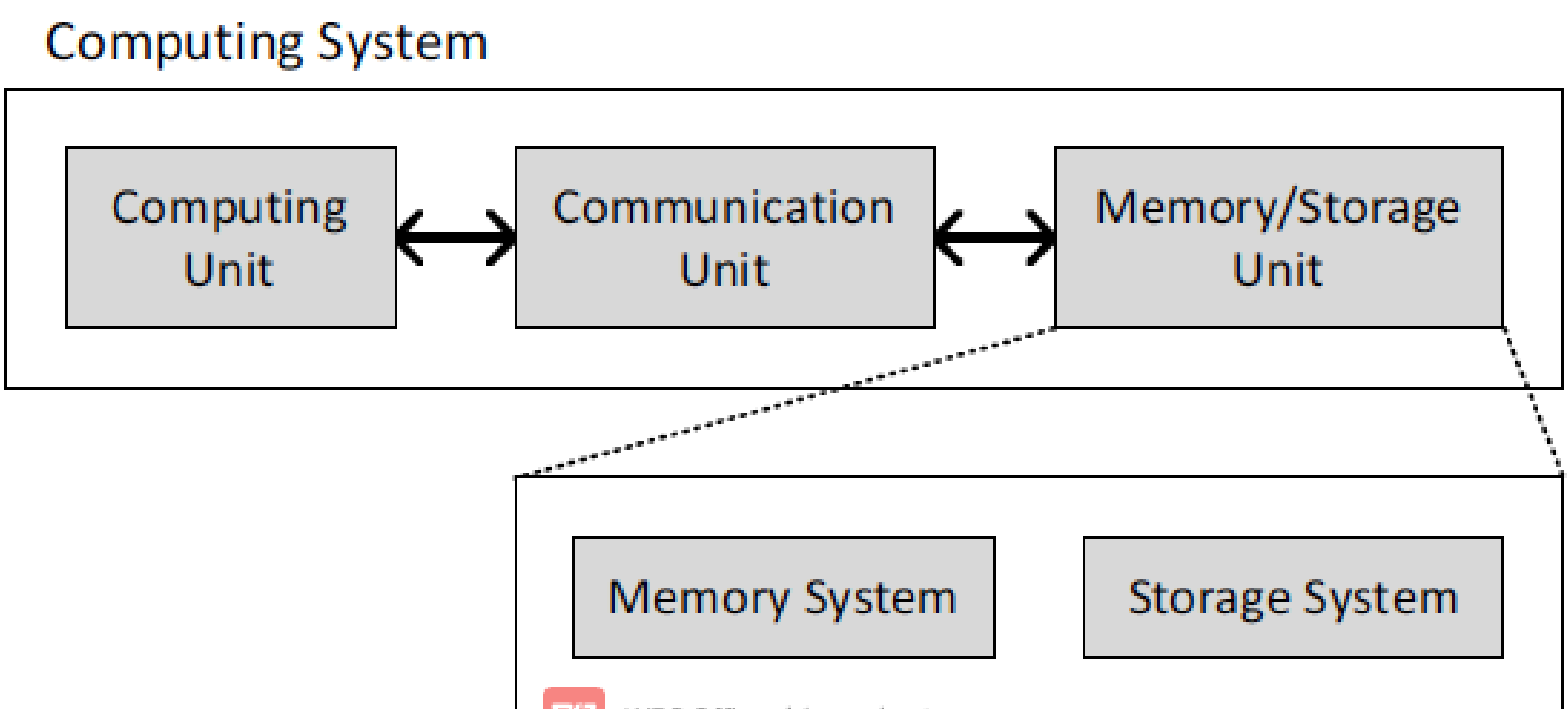
- You can change the world only if you understand it well enough…
  - Especially the past and present dominant paradigms
  - And, their advantages and shortcomings -- tradeoffs

INSTRUCTION SET ARCHITECTURE (ISA)

What is A Computer?

- Three key components

- Computation
- Communication
- Storage (memory)



The Von Neumann Model/Architecture

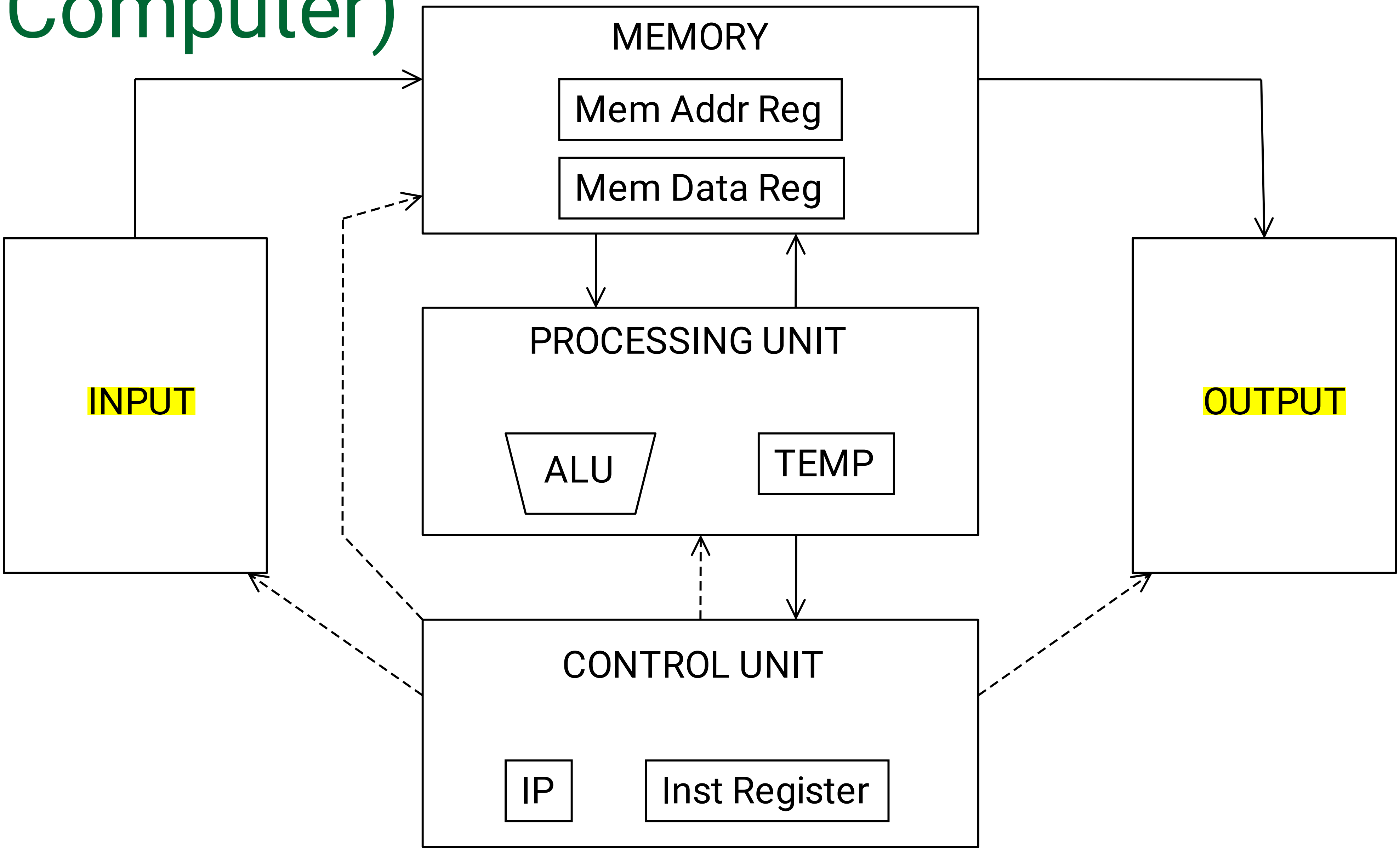
- Also called *stored program computer* (instructions in memory). Two key properties:

- Stored program
  - Instructions stored in a linear memory array
  - Memory is unified between instructions and data
    - The interpretation of a stored value depends on the control signals

When is a value interpreted as an instruction?

- Sequential instruction processing
  - One instruction processed (fetched, executed, and completed) at a time
  - Program counter (instruction pointer) identifies the current instr.
  - Program counter is advanced sequentially except for control transfer instructions

The von Neumann Model (of a Computer)



The Dataflow Model (of a Computer)

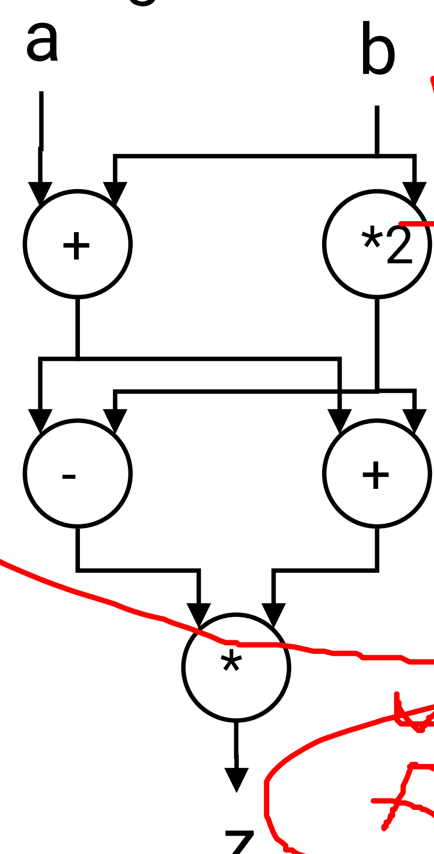
- Von Neumann model: An instruction is fetched and executed in control flow order
  - As specified by the instruction pointer
  - Sequential unless explicit control flow instruction

- Dataflow model: An instruction is fetched and executed in data flow order
  - i.e., when its operands are ready
  - i.e., there is no instruction pointer
  - Instruction ordering specified by data flow dependence
    - Each instruction specifies “who” should receive the result
    - An instruction can “fire” whenever all operands are received
  - Potentially many instructions can execute at the same time
    - Inherently more parallel

von Neumann vs Dataflow

- Consider a von Neumann program
  - What is the significance of the program order?
  - What is the significance of the storage locations?

```
v <= a + b;
w <= b * 2;
x <= v - w
y <= v + w
z <= x * y
```



Sequential

von numenn model

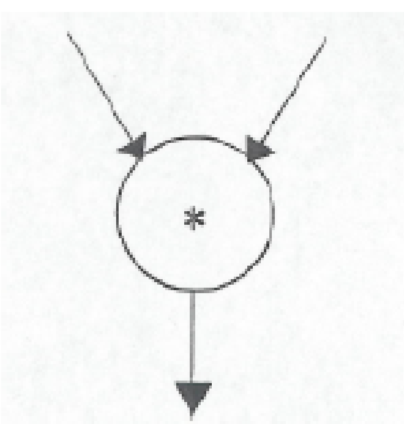
Dataflow

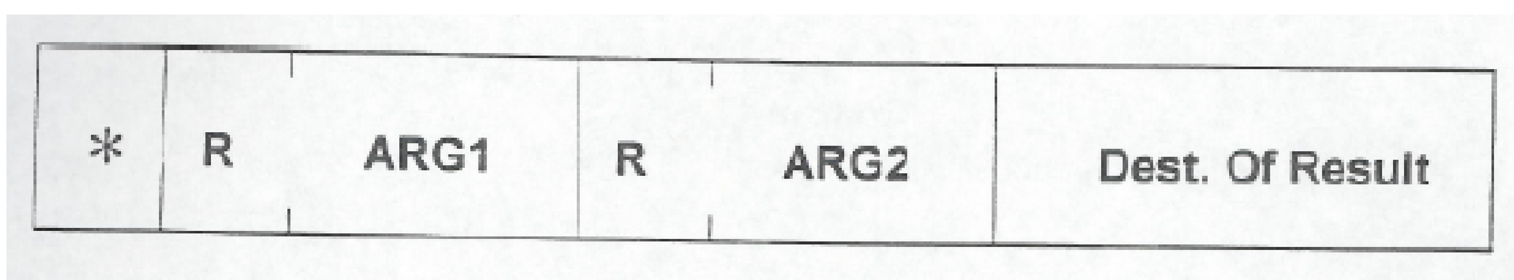
- Which model is more natural to you as a programmer?

More on Data Flow

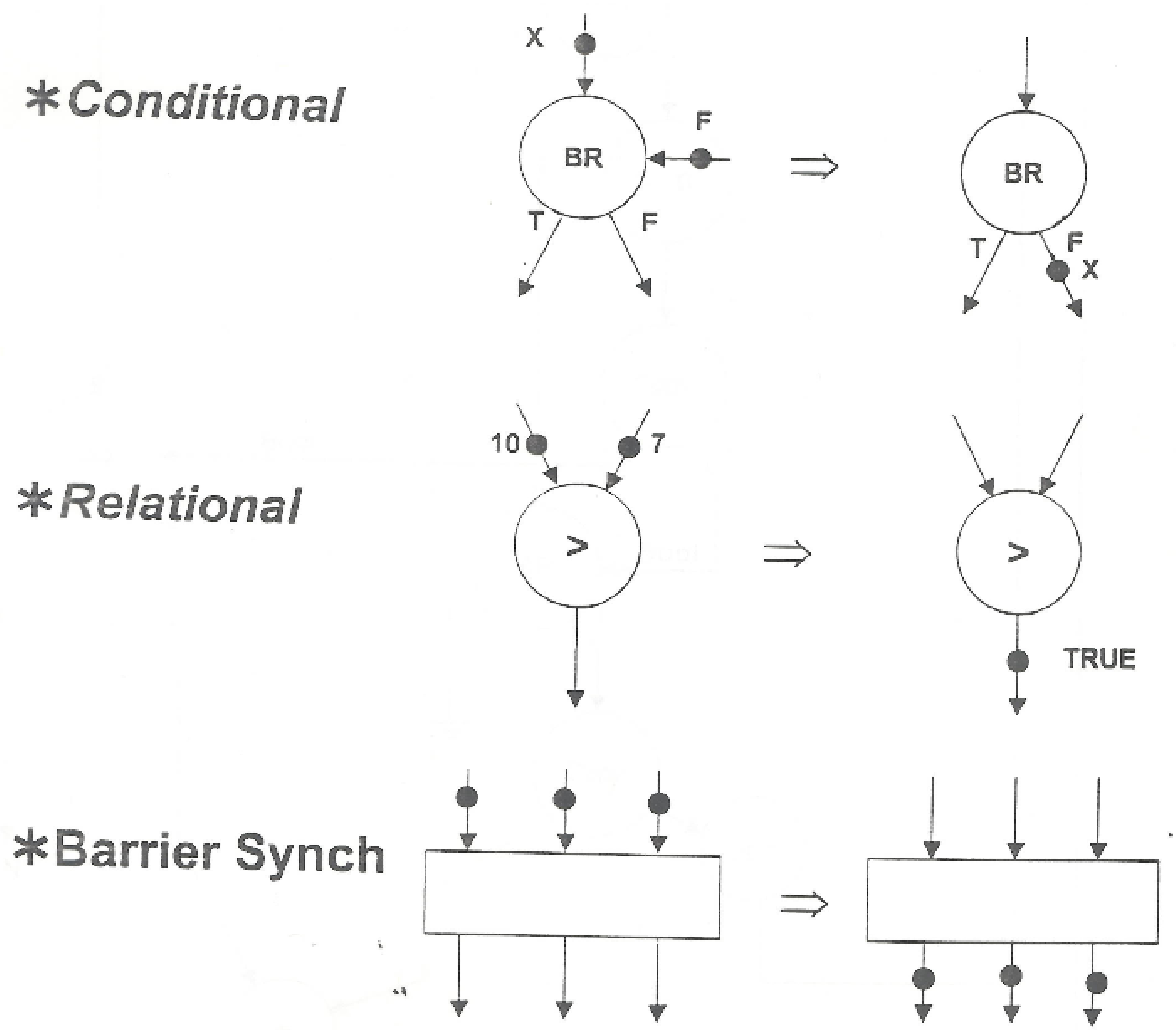
- In a data flow machine, a program consists of data flow nodes
  - A data flow node fires (fetched and executed) when all it inputs are ready
    - i.e. when all inputs have tokens

- Data flow node and its ISA representation

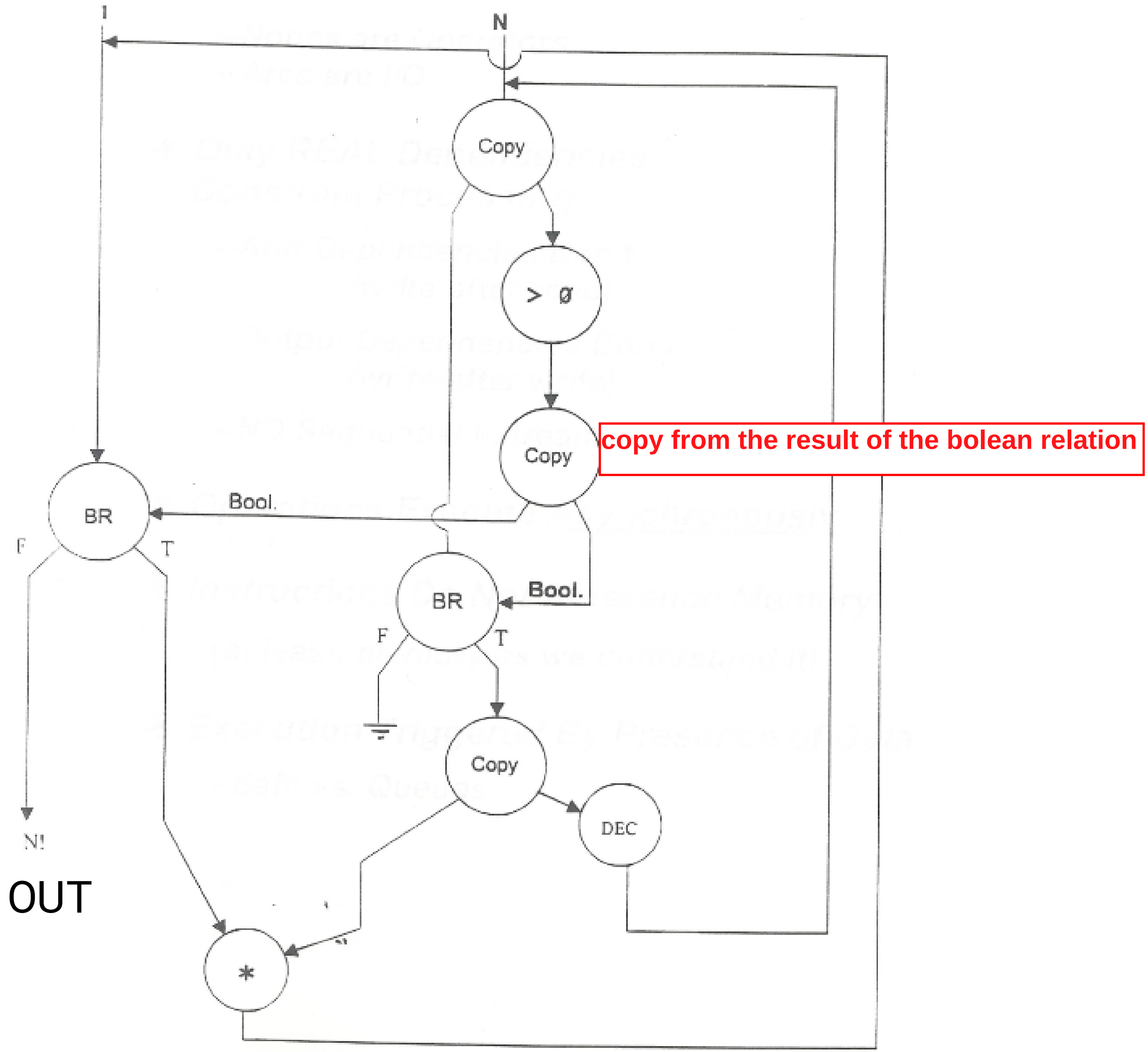




Data Flow Nodes



An Example Data Flow Program



ISA-level Tradeoff: Instruction Pointer

- Do we need an instruction pointer in the ISA?
  - Yes: Control-driven, sequential execution
    - An instruction is executed when the IP points to it
    - IP automatically changes sequentially (except for control flow instructions)
  - No: Data-driven, parallel execution
    - An instruction is executed when all its operand values are available (data flow)

- Tradeoffs: MANY high-level ones
  - Ease of programming (for average programmers)?
  - Ease of compilation?
  - Performance: Extraction of parallelism?
  - Hardware complexity?

Remember

- Von Neumann and Data-flow is just models

- All major instruction set architectures today use this Von Neumann model
    - x86, ARM, MIPS, SPARC, Alpha, POWER

MICROARCHITECTURE

ISA vs. Microarchitecture

- What is part of ISA vs. Uarch?
  - Gas pedal: interface for “acceleration”
  - Internals of the engine: implement “acceleration”
- ISA
  - Agreed upon interface between software and hardware (SW/compiler assumes, HW promises)
  - What programmer needs to know to write and debug system/user programs and to specify to user a sequential control-flow or a data-flow execution order.
- Microarchitecture
  - Specific implementation of an ISA
  - Not visible to the software or the programmer
- Microprocessor
  - **ISA, uarch**, circuits
  - “Architecture” = ISA + microarchitecture

Problem
Algorithm
Program
ISA
Microarchitecture
Circuits
Electrons

ISA Example

- Instructions
  - Opcodes, Addressing Modes, Data Types
  - Instruction Types and Formats
  - Registers, Condition Codes
- Memory
  - Address space, Addressability, Alignment
  - Virtual memory management
- Call, Interrupt/Exception Handling
- Access Control, Priority/Privilege
- I/O: memory-mapped vs. instr.
- Task/thread Management
- Power and Thermal Management
- Multi-threading support, Multiprocessor support

Intel® 64 and IA-32 Architectures Software Developer's Manual

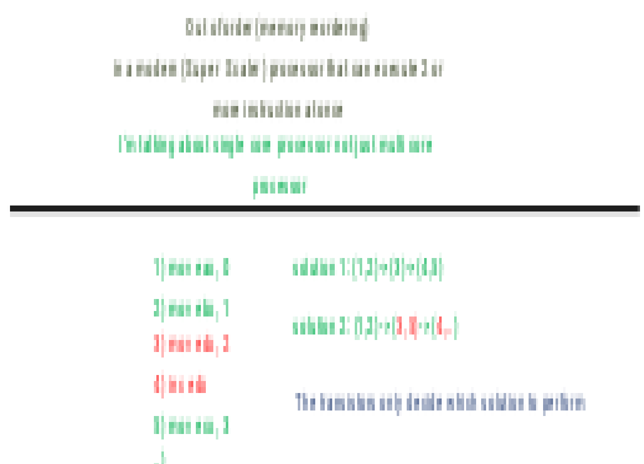
Microarchitecture Example

- Underneath (at the microarchitecture level), the execution model of almost all *implementations (or, microarchitectures)* is very different
  - Pipelined instruction execution: *Intel 80486 uarch*
  - Multiple instructions at a time: *Intel Pentium uarch*
  - Out-of-order execution: *Intel Pentium Pro uarch*
  - Separate instruction and data caches
- Implementation (uarch) can be various as long as it satisfies the specification (ISA)
  - Add instruction vs. Adder implementation
    - Bit serial, ripple carry, carry lookahead adders are all part of microarchitecture
  - x86 ISA has many implementations: 286, 386, 486, Pentium, Pentium Pro, Pentium 4, Core, …

Microarchitecture Tradeoff: control vs. data driven

- A similar tradeoff (control vs. data-driven execution) can be made at the microarchitecture level

- Microarchitecture: How the underlying implementation actually executes instructions
  - Microarchitecture can execute instructions in any order as long as it obeys the semantics specified by the ISA when making the instruction results visible to software
    - Programmer should see the order specified by the ISA



Review Questions (ISA or Uarch)?

- ADD instruction's opcode **ISA**
- Number of general purpose registers **ISA**
- Number of ports to the register file **Uarch**
- Number of cycles to execute the MUL instruction **Uarch**
- Whether or not the machine employs pipelined instruction execution **Uarch**

Remember

- Microarchitecture: Implementation of the ISA under specific design constraints and goals

- Microarchitecture usually changes faster than ISA
  - Few ISAs (x86, ARM, SPARC, MIPS, Alpha) but many uarchs
  - *Why?*
    - *Abstraction!*

COMPUTER ARCHITECTURE

What is Computer Architecture?

- **ISA+implementation definition:** The science and art of designing, selecting, and interconnecting hardware components and designing the hardware/software interface to create a computing system that meets functional, performance, energy consumption, cost, and other specific goals.

- **Traditional (only ISA) definition:** "The term *architecture* is used here to describe the attributes of a system as seen by the programmer, i.e., the conceptual structure and functional behavior as distinct from the organization of the dataflow and controls, the logic design, and the physical implementation." *Gene Amdahl*, IBM Journal of R&D, April 1964

Next lecture.

Design Point

- A set of design considerations and their importance
  - leads to tradeoffs in both ISA and uarch
- Considerations
  - Cost
  - Performance
  - Maximum power consumption
  - Energy consumption (battery life)
  - Availability
  - Reliability and Correctness
  - Time to Market

Problem
Algorithm
Program
ISA
Microarchitecture
Circuits
Electrons

- Design point determined by the "Problem" space (application space), or the intended users/*market*

Application Space

- Dream, and they will appear…

Other examples of the application space that continue to drive the need for unique design points are the following:

1) scientific applications such as those whose computations control nuclear power plants, determine where to drill for oil, and predict the weather;
2) transaction-based applications such as those that handle ATM transfers and e-commerce business;
3) business data processing applications, such as those that handle inventory control, payrolls, IRS activity, and various personnel record keeping, whether the personnel are employees, students, or voters;
4) network applications, such as high-speed routing of Internet packets, that enable the connection of your home system to take advantage of the Internet;
5) guaranteed delivery (a.k.a. real time) applications that require the result of a computation by a certain critical deadline;
6) embedded applications, where the processor is a component of a larger system that is used to solve the (usually) dedicated application;
7) media applications such as those that decode video and audio files;
8) random software packages that desktop users would like to run on their PCs.

Each of these application areas has a very different set of characteristics. Each application area demands a different set of tradeoffs to be made in specifying the microprocessor to do the job.

Tradeoffs: Soul of Computer Architecture

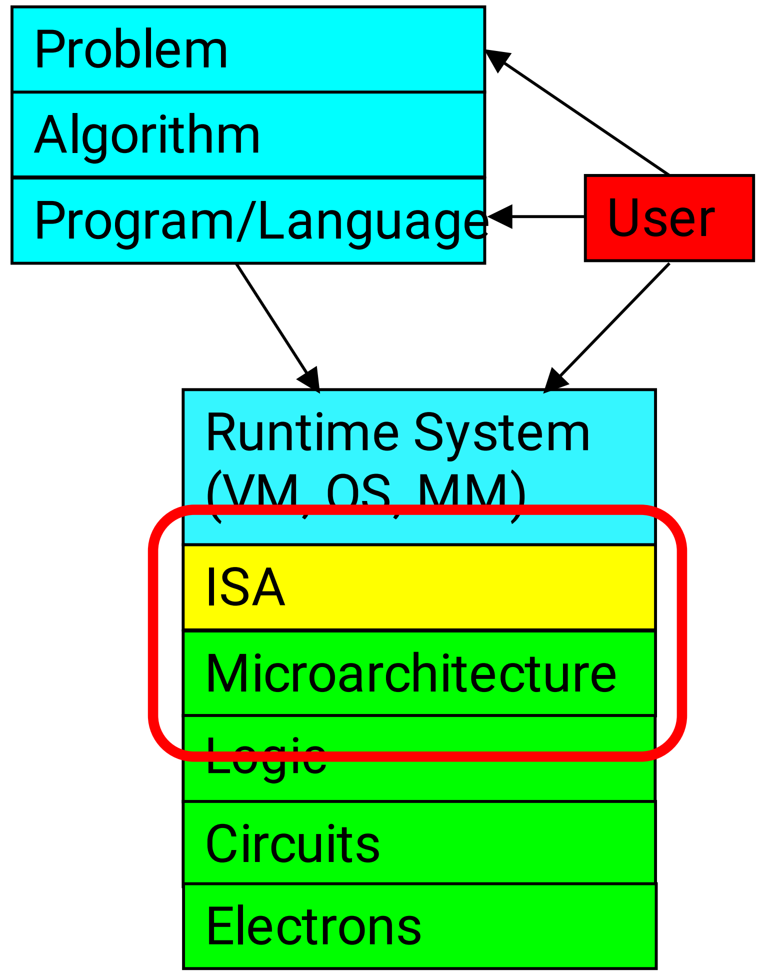
- ISA-level tradeoffs

- Microarchitecture-level tradeoffs

- System and Task-level tradeoffs
  - How to divide the labor between hardware and software

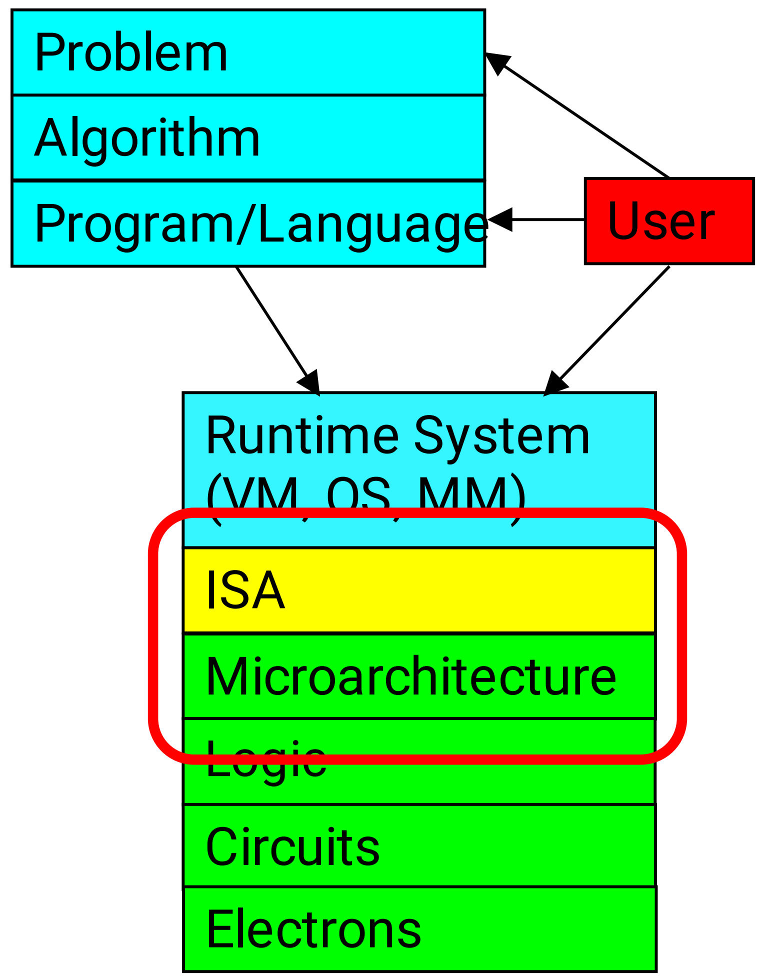
- *Computer architecture is the science and art of making the appropriate trade-offs to meet a design point*
  - *Why art?*

Why Is It (Somewhat) Art?



- We do not (fully) know the future (applications, users, market)

Why Is It (Somewhat) Art?



- And, the future is not constant (it changes)!

Analog from Macro-Architecture

- Future is not constant in macro-architecture, either
- Example: Can a power plant boiler room be later used as a classroom?

Macro-Architecture: Boiler Room

At the west end of campus was a small structure that housed the boiler room that functioned as the school's power plant. Below, in the rain beside the railroad tracks, a farmer's goat grazed and occasionally wandered up to eat the grass of this yet untamed end of campus.

Over a 20 month period from 1912 - 1914, Machinery Hall was built on top of that boiler room. The massive tower, which has become a symbol of Carnegie Mellon, was designed to disguise the smokestack. Architect Henry Hornbostel had created a "temple of technology" that would become one of the most renowned buildings of the Beaux Arts style in the country.

Early course catalogs described the boiler room as a classroom where students learned about power generating machinery. The tower continued to belch smoke until 1975, but in 1979 the boiler room became the cleanest room on campus with the construction of the Nanofabrication Facility. The coal bin area became the offices and computer room of the D-level.